Maintenance Procedure — KP-40 Hydraulic Pump

Document KP-MP-012, revision B, approved by the workshop manager.

Purpose

This procedure describes the periodic maintenance of the KP-40 hydraulic pump installed on the test benches of buildings C and D.

It applies to every technician who has completed the basic hydraulics training and holds a valid workshop badge.

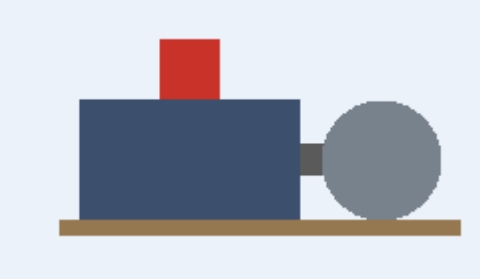


Figure 1: the KP-40 hydraulic pump.

It replaces revision A, which no longer applies after 1 December 2026.

Scope

The following operations are covered:

- Visual inspection of the pump housing, hoses and fittings
- Replacement of the suction filter
- Check of the oil level and color
- Measurement of the outlet pressure
- Cleaning of the cooling fins

Repairs of the motor and of the electrical cabinet are not covered by this document; they are described in procedure KP-MP-020.

Procedure

1. Stop the pump and lock out the main switch.
2. Place a drip tray under the filter housing.
3. Open the drain valve and let the oil flow into the tray.
4. Unscrew the filter cover with the KP-T-05 strap wrench.

5. Remove the old filter element and discard it with the oily waste.
6. Fit a new element, reference KP-F-110, and screw the cover back and tighten it to 25 N·m.
7. Restart the pump, wait two minutes and check that no leak appears around the cover.

The parts needed for one maintenance are listed below.

Part	Reference	Quantity
Suction filter element	KP-F-110	1
Cover seal	KP-J-031	1
Drain valve seal	KP-J-034	1
Hydraulic oil ISO VG 46	KP-H-046	3 litres

Safety

Before any operation, the pump must be stopped, isolated and locked out with a personal padlock.

The residual pressure in the circuit can reach 15 bar for several minutes after the stop[1]; wait at least fifteen minutes before opening any connection.

Wear safety glasses and oil-resistant gloves at all times.

[1] Measured on the benches of buildings C and D in March 2026.

Records

Record the date, the outlet pressure and your name in the maintenance log kept ~~next to the bench~~in the workshop office. A pressure below 180 bar must be reported to the workshop manager the same day.